10
Spacecraft Contamination

10.1 Introduction

This chapter addresses the potential contamination of spacecraft systems by molecular, particulate, and biological materials. Contamination of spacecraft surfaces by deposition of molecular and particulate matter can significantly change their properties such that the mission could be adversely affected. Contamination may be in the form of a liquid, gas, or solid and may be uniformly or non-uniformly distributed on surfaces and in liquids or gases. Contamination can occur during subsystem development, spacecraft integration, subsystem and system testing, launch, and in-orbit operations.

Contamination generally occurs by absorption or adsorption. *Absorption* is the process by which one material assimilates into another, while *adsorption* is the adhesion of molecules to a solid surface. *Desorption* is the process of removing adsorbed or absorbed substances from a solid or liquid. In addition to the contaminants on the surface of a material, gaseous atoms may be dissolved in a bulk medium whose distribution is regulated by diffusion. *Diffusion* is the process whereby atoms or molecules move from a region of high concentration to a region of low concentration. Atoms or molecules that diffuse and migrate toward the surface in contact with the vacuum of space will then desorb.

The deposition of molecular species on critical surfaces can lead to changes in thermal and optical properties, such as transmittance, reflectance, and absorptance. Contamination of a thermal control surface could result in the spacecraft or instrument being unable to maintain its proper temperature, which may adversely affect performance. Contamination of an optical system could result in serious performance degradation, such as poor imaging and obscuration of selected frequencies. In addition, contaminants in the environment in the vicinity of optical sensors could produce scattering and distortion of images. Contamination of exterior surfaces of a spacecraft could also produce interactions with the environment, resulting in surface reactions. As a consequence, each subsystem on a spacecraft should be analyzed 1) for its sensitivity to contamination, 2) as a potential source of contamination, 3) for its potential degree of contamination, and 4) for potential degradation of performance due to contamination.

A *contamination control plan* should be instituted at the beginning of an instrument or spacecraft development project. The role of contamination control engineering is to improve the performance of the system by reducing contamination to an acceptable level. It may include 1) establishing a contamination control program of procedures and facilities to reduce contamination to an acceptable level during development, test, transportation, launch, and in-orbit operation; 2) selecting materials and coatings; 3) analyzing the design for contamination issues using various modeling techniques; 4) designing, developing, and/or utilizing sensor systems to monitor contamination during development, test, transportation, launch, and operation; 5) assuring that facilities used satisfy the contamination control requirements; and 6) assisting in the in-orbit operations to minimize in-orbit contamination.

Planetary protection is concerned with the contamination of other bodies in the solar system with microbes from Earth that might interfere with scientific exploration and irreversibly change their environment. There is also concern that spacecraft returning from other planetary environments could return microbes that could harm humans and the ecosystem on Earth. In this regard, the Outer Space Treaty, which represents the basic legal framework on international space law signed in 1967, includes the following statement on planetary protection in Article IX:

> States Parties to the Treaty shall pursue studies of outer space, including the Moon and other celestial bodies, and conduct exploration of them so as to avoid their harmful contamination and also adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter and, where necessary, shall adopt appropriate measures for this purpose.

Quantitative assessment and modeling of spacecraft contamination is complex due to the large numbers of materials generally involved, the variability of the characteristics of the contaminants, interactions of the contaminants with surfaces, residual gases in the vicinity of the spacecraft, the neutral and ionized spacecraft ambient environment, and the variation in temperature as the spacecraft travels through the sunlight and the umbra of the Earth or another celestial body.

10.2 Material Outgassing

Outgassing is the release of gases by a material over time, and the rate of outgassing is generally exacerbated by exposure to the near vacuum of space. Historically, only materials that meet the criteria of ASTM E-595-93(2003)e2 (American Society for Testing and Materials 2003a), i.e., that have a total mass loss (TML) <1.0% and a *collected volatile condensable mass* (CVCM) <0.10%, are approved for use in the space environment unless application considerations dictate otherwise. The %TML is the amount of material that outgasses from a conditioned sample, expressed as a percent of the initial sample mass. If a sample with a mass of 100 g loses 1 g of mass during the test, the %TML is 1%. In some instances the total mass lost will include absorbed water, which in moderate quantities will dissipate in the vacuum of space and not be problematic. The %CVCM represents the percentage of the mass that outgasses from the sample and then condenses on a collector plate. This test is a measure of what might condense on critical surfaces such as optics and thermal control surfaces that are often at lower temperatures. A low %CVCM indicates that a small percentage of the mass loss is other than water vapor. The percent of water vapor regain (%WVR) is a measure of how much water is reabsorbed by the material after being reexposed to a humid environment. This criterion is not a space requirement but provides insight into the need for desiccation and/or drying of materials during the fabrication process.

The ASTM E-595-93(2003)e2 standard (American Society for Testing and Materials 2003a) describes the test method. The %TML is the mass loss while the material is heated to 125°C at a pressure less than 7×10^{-3} Pa (5×10^{-5} torr) for 24 h. The %CVCM is the percentage of the mass that condenses in a container maintained at a temperature of 25°C while the %TML test is carried out. The %WVR is a measure of how much water will be reabsorbed by the material after being reexposed to an environment of 50% relative humidity at 23°C for 24 h.

Because outgassing is such an important consideration in spacecraft design, NASA maintains an online database of material outgassing properties (Campbell and Scialdone 2006). This database contains materials grouped in 18 categories according to their primary use, such as adhesives, conformal coatings, tapes, marking materials, etc. The database consists of the name of each material and its components, TML, CVCM, cure time, cure temperature, atmosphere, WVR, data reference, and application. Selected entries from the database are in Table 10.1.

A test for the kinetics of the outgassing from a material is specified in ASTM E1559-03 (American Society for Testing and Materials 2003b), which is beyond the scope of this treatment. This test method can be used to produce the data necessary to support mathematical models used for the prediction of molecular contaminant generation, migration, and deposition. The outgassing rate $\dot{Q}$ can be measured by placing a sample with a given surface area (m^2) into a vacuum chamber at a given temperature and known volume (m^3) and having the pressure increase (Pa) over an interval of time (s). The temperature used is generally room temperature at 21–23°C. As a result, the units for $\dot{Q}$ are (pressure increase) $\times$ (volume of chamber) $\times$ (area of sample)$^{-1}$ $\times$ (time in chamber)$^{-1}$ at a specified temperature, Pa $m^3\ m^{-2}\ s^{-1}$, which is equivalent to the SI unit for outgassing of W m^{-2}. The values in pressure volume units can be transformed to number of molecules by dividing by kT using the perfect gas law $P = NkT$, where N is the number density and T is the temperature. As a result, the outgassing rate $\dot{Q}$ (W m^{-2}) can be expressed in terms of N outgassing molecules per unit area per unit time (molecules $m^{-2}\ s^{-1}$) as

$$\dot{N} = \frac{\dot{Q}}{kT} \tag{10.1}$$

so that $\dot{m}$, the outgassing mass per unit area per unit time (kg $m^{-2}\ s^{-1}$), follows as

$$\dot{m} = \frac{M\dot{N}}{N_A} = \frac{\dot{Q}M}{kTN_A} \tag{10.2}$$

where

N_A = Avogadro's number, particles $kmol^{-1}$
K = Boltzmann's constant, J K^{-1}
M = molecular mass, kg $kmol^{-1}$
T = temperature, K

Table 10.1 Sample outgassing data for selected spacecraft materials

Material	%TML	%CVCM	Cure time	Cure temperature, K	Atmosphere	%WVR	Data reference	Application
EC 2258 epoxy	1.00	0	4 h	65	Air		GSFC3371	Adhesive
PS 869 clear epoxy	0.84	0.01	24 h	25	Air	0.2	GSC23317	Adhesive
Scotchweld 1838	0.96	0.05	7 days	25	Air	0.22	GSC23547	Adhesive
Arathane (uralane) 5750	0.78	0.02	7 days	25	Air	0.07	GSC28807	Conformal coating
Conathane EN-11 thinned /sprayed/F	0.62	0.08	16 h	25	Air	0.14	GSC14596	Conformal coating
Epoxy 20-3651/Cat 105 As A/B 100/8 Bw Black	0.50	0.03	1 h	120	Air	0.12	GSC23484	Potting
Amphenol connector 97 series 99-825-730-15 Ultem	0.73	0.02				0.27	GSC27166	Connector
Bendix gasket MS27656T25F61SJO31160 red silicone	0.11	0.05	24 h	190	10^{-2} torr	0.01	GSC14882	Connector gasket
Conn Insul Mil-C38999 Parker 5899-50 red silicone	0.18	0.10	24 h	149	10^{-2} torr	0.03	GSC20172	O ring
Aeroglaze Z306 black top-coat/9929 epoxy primer/F	0.90	0.01	2 days	60	10^{-6} torr	0.34	GSC23799	Paint composite
BE 620 white reflector paint	0.56	0.02	7 days	25	Air	0.11	GSC18546	Paint
Chemglaze Z306 lot Cja/F	1.00	0.02	40 days	25	Air	0.57	GSC14442	Thermal control
Duocel aluminum foam 10 Ppi 6–8% density	0.03	0.00				0.01	GSC14934	Structural
Min-K 1301 Ht insulation	0.51	0.06					GSFC3329	Insulation
Velcro midtemp fastener nomex/polyimide	0.52	0.00					GSFC3883	Fastener

Source: From Campbell and Scialdone (2006); courtesy of NASA.

Table 10.2 Conversion of non-SI outgassing units

To convert entry to W m^{-2}	Multiply by
Pa L m^{-2} s^{-1}	1×10^{-3}
torr L cm^{-2} s^{-1}	1.3332×10^{3}
molecules m^{-2} s^{-1} at 296 K	2.4470×10^{20}
kmoles m^{-2} s^{-1} at 296 K	4.0633×10^{-7}

Conversion of the outgassing rate $\dot{Q}$ in non-SI units to SI units is possible by the conversion factors given in Table 10.2. Typical outgassing rates of selected materials are given in Table 10.3.

10.3 Cleanliness Levels

Cleanliness levels are defined by the standard IEST-STD-CC1246D (Institute of Environmental Sciences and Technology 2002), which is a revision of the standard that was first published as MIL-STD-1246 (U.S. Department of Defense 1962). This standard defines quantitative cleanliness levels for items that consist of solid and fluid (gas or liquid) components. Contaminants are usually separated into two types, particles and nonvolatile residue (NVR). *Particles* or *particulates* are defined as small solid or liquid material with definable shape or mass. *NVR* refers to the quantity of molecular and particulate matter remaining following the evaporation of a solvent containing the contaminants. The standard defines separate levels of cleanliness for particles and NVR. Only particulates of sizes greater than 1 μm and NVR levels greater than 1 ng per 0.1 liter are deemed significant. Based on experimental data on deposition of particulates on surfaces, it was found that the distribution of the number of particulates greater than a given size could be represented by a log normal distribution with a minimum size of 1 μm, where

$$\log_{10} N_x = 0.926(\log_{10}^2 L - \log_{10}^2 x) \tag{10.3}$$

Table 10.3 Typical outgassing rates

Material	Treatment	Outgassing rate, W m^{-2}
Aluminum	None	1×10^{-3}
Aluminum	Degassed	2×10^{-4}
Aluminum 6061-T6	None	3×10^{-6}
Aluminum 6061-T6	Baked at 200°C	6×10^{-6}
Kapton foil		1×10^{-4}
Polyimide	Baked at 300°C	5×10^{-5}
Nylon		8×10^{-3}
Teflon		3×10^{-5}

where

N_x = number of particles per 0.1m^2 greater than x
L = cleanliness level
x = maximum linear dimension of a particle, μm

The original definition for N_x was with respect to an area in ft^2, but the units were changed to 0.1 m^2 with the differences between them deemed insignificant. Cleanliness levels defined by Eq. (10.3) are given in Table 10.4 and Fig. 10.1.

Table 10.5 gives the categories for levels of nonvolatile residue.

10.4 Clean Room Cleanliness

Classification of air cleanliness in clean rooms and associated controlled environments is defined by the standard ISO-14644-1 (International Organization for Standardization 1999). The classifications are defined solely in terms of the concentration of airborne particulate contamination to levels appropriate for accomplishing contamination-sensitive activities. The maximum number of particles permitted for any given size is given by

$$C_n = 10^n \left(\frac{0.1}{D}\right)^{2.08} \tag{10.4}$$

where

C_n = maximum permitted number of particles per unit volume with particle size $\geq D$, rounded to a whole number, particles m^{-3}
n = ISO class number from 1 to 9
D = particle size with particle specification of multiples of 0.1 μm

The maximum number of particles greater than and equal to a specified size for a given class is given in Table 10.6 and Fig. 10.2. As a comparison, the sizes of typical particles are illustrated in Table 10.7.

To verify particle dispersion, it is necessary to collect deposited particles on a witness plate, a clean flat object that is known to be free of contaminants and is similar in material characteristics to the item of interest. Several witness plates should be dispersed throughout the clean room, and the particles collected on them should be counted periodically. Counting and sizing can be done by an optical microscope or a laser optical surface particle counter.

Positive pressure clean rooms are used when the air escaping is not hazardous. Usually, there is a decrease in positive pressure between the clean room and outer rooms, such as a gowning room, which is maintained at a lower positive pressure. The positive pressure is generally maintained at 0.05–0.5 cm of water relative to the surroundings.

10.5 Contamination Control Program

Spacecraft cleanliness levels must be maintained throughout fabrication, integration, subsystem- and system-level testing, launch integration, launch, and

Table 10.4 Particulate cleanliness levels, from Eq. (10.3)

Level	Reference particle size, μm	Maximum allowed limits for stated size and larger particles, per 0.1 m^2 surface area, or per 0.1 liter of gas or fluid
1	1	1.0
5	1	2.8
5	2	2.3
5	5	1.0
10	1	8.4
10	2	6.9
10	5	3.0
10	10	1.0
25	2	53.1
25	5	22.7
25	15	3.4
25	25	1.0
50	5	166
50	15	24.7
50	25	7.3
50	50	1.0
100	5	1784
100	15	265
100	25	78.4
100	50	10.7
100	100	1.0
200	15	4188
200	25	1240
200	50	170
200	100	15.8
200	200	1.0
300	25	7454
300	50	1021
300	100	95
300	250	2.3
300	300	1.0
500	50	11816
500	100	1100
500	250	26.4
500	500	1.0
750	50	95806
750	100	8919
750	250	214
750	500	8.1
750	750	1.0
1000	100	42658
1000	250	1022
1000	500	38.8
1000	750	4.8
1000	1000	1.0

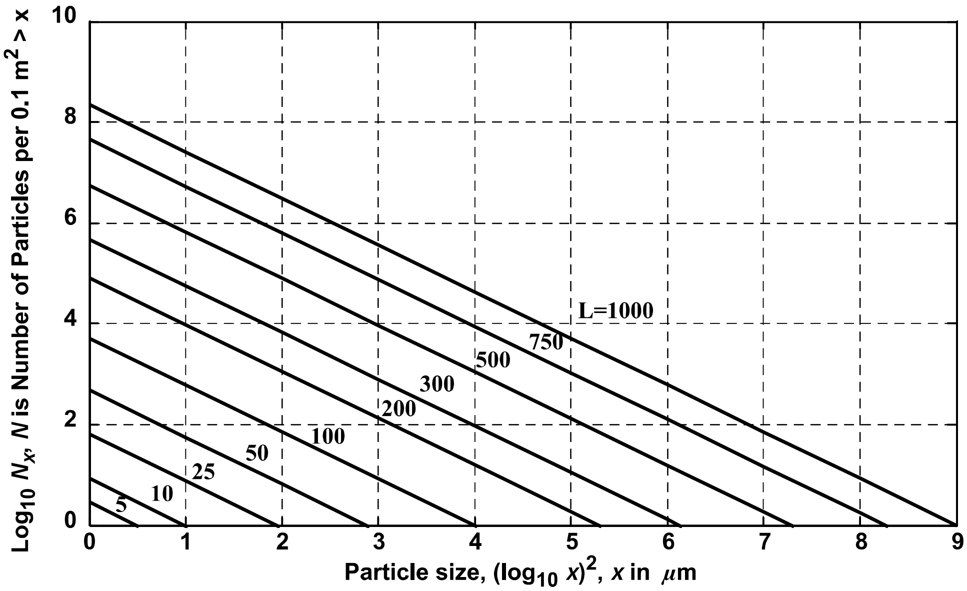

Fig. 10.1 Particle cleanliness level where L is cleanliness level, from Eq. (10.3).

operations. To assure cleanliness, a contamination control program should be instituted for all spacecraft projects that have stringent cleanliness requirements; see IEST-STD-CC1246D (Institute of Environmental Sciences and Technology 2002) and PD-ED-1233 [NASA (undated)]. The program should establish cleanliness specifications at the beginning of the design phase. The most stringent

Table 10.5 Limits of nonvolatile residue (NVR) for surface, liquid, or gas

Level	Limit NVR, μg cm^{-2} or μg per 0.1 liter
AA5	0.001
A/100	0.01
A/50	0.02
A/20	0.05
A/10	0.1
A/5	0.2
A/2	0.5
A	1.0
B	2.0
C	3.0
D	4.0
E	5.0
F	7.0
G	10.0
H	15.0
J	25.0

Source: From IEST-STD-CC1246D (Institute of Environmental Sciences and Technology 2002).

Table 10.6 Cleanliness classification, from Eq. (10.4)

ISO class	Maximum number of particles per cubic meter ≥ specified size					
	$D = 0.1\ \mu m$	$D = 0.2\ \mu m$	$D = 0.3\ \mu m$	$D = 0.5\ \mu m$	$D = 1\ \mu m$	$D = 5\ \mu m$
1	10	2				
2	100	24	10	4		
3	1,000	237	102	35	8	
4	10,000	2,370	1,020	352	83	
5	100,000	23,700	10,200	3,520	832	29
6	1,000,000	237,000	102,000	35,200	8,320	293
7				352,000	83,200	2,930
8				3,520,000	832,000	29,300
9				35,200,000	8,320,000	293,000

specifications are generally established for cold surfaces such as required for optical instruments, solar array surfaces, and thermal control surfaces. Utilization of special contamination control domes is generally employed to protect critical surfaces during integration, subsystem- and system-level testing, transportation, and launch integration. In some cases, they are retained during launch and deployed in orbit after there has been sufficient outgassing of the spacecraft.

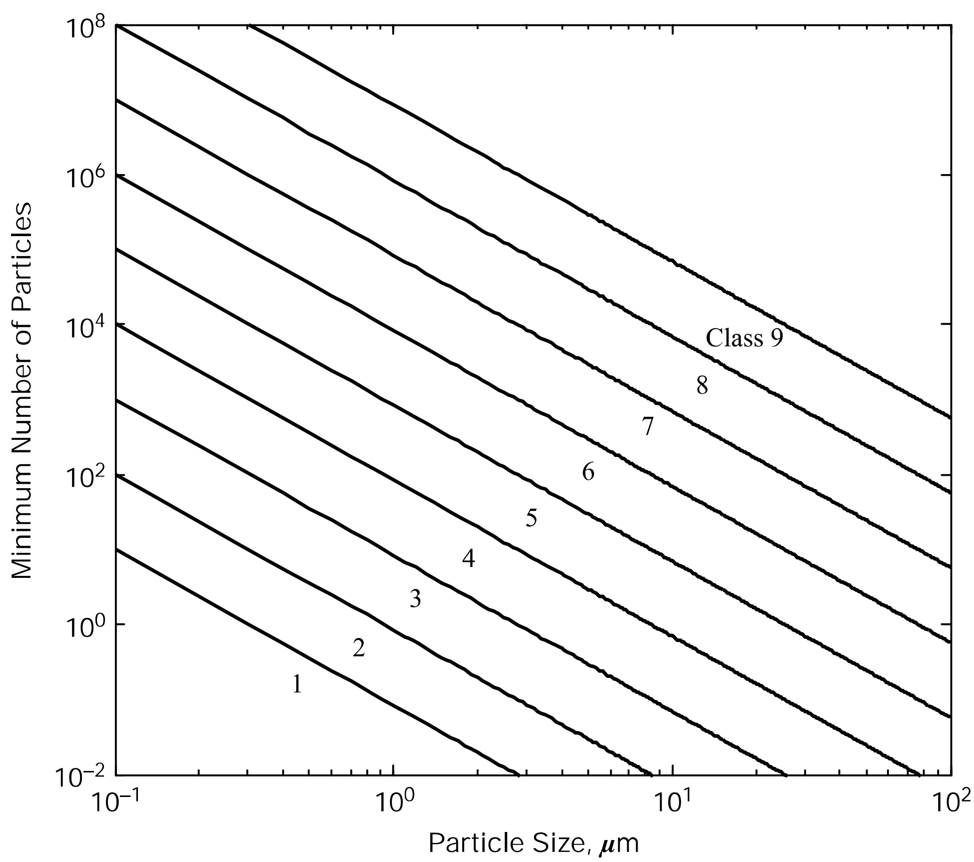

Fig. 10.2 Maximum particle sizes for clean room classes, from Eq. (10.4).

Table 10.7 Typical particle sizes

Particle	Approximate size, μm
Human hair diameter	50 – 150
Metallurgical dust and fumes	0.1 – 5
Flu virus	0.7
Pollen	7.0 – 100
Sneeze particles	10 – 300

10.6 Contamination Analysis

10.6.1 Outgassing Rates

The extent of outgassing of a material in a low-pressure environment is characterized by 1) the specific material, 2) the environment to which the material was previously exposed, 3) increasing significantly if the temperature is increased, 4) decreasing as a function of time, and 5) being a relatively weak function of the environmental pressure. The time dependence of outgassing depends on the specific process and the specific materials involved. Material that outgasses at a rate proportional to the mass available can be modeled by a first-order reaction, where

$$\frac{\mathrm{d}m_s}{\mathrm{d}t} = -m_s \tau_s^{-1} \tag{10.5}$$

where

m_s = mass of the contaminant s per unit area, kg m^{-2}
s = identifies a particular species of contaminant
t = time, s
τ_s = residence time, s

Residence time is defined as the time that a particle spends in a particular location. For a constant τ_s, it follows from Eq. (10.5) that

$$m_s = m_{s,0}\, e^{-t/\tau_s} \tag{10.6}$$

where $m_{s,0}$ is the initial mass of contaminant s per unit area (kg m^{-2}). The mass loss follows directly as

$$m_{s,\mathrm{loss}} = m_{s,0}\,(1 - e^{-t/\tau_s}) \tag{10.7}$$

The residence time can be determined from the Arrhenius equation for the temperature dependence of a chemical reaction as

$$\tau_s = \tau_{s,0}\, e^{-E_{s,a}/RT} \tag{10.8}$$

where

$E_{s,a}$ = the activation energy for desorption of contaminant s, J kmol^{-1}
R = the universal gas constant, J kmol^{-1} K^{-1}
T = temperature, K
$\tau_{s,0}$ = resident time at zero temperature Celsius, s

The *activation energy* is the threshold energy, or the energy that must be overcome in order for a chemical reaction to occur. The *activation energy for desorption* is the threshold energy reguired for desorption to occur. The resident time at nominal temperature $\tau_{s,0}$ is estimated from

$$\tau_{s,0} \approx \frac{h}{kT} \approx 1.7 \times 10^{-13}\,\mathrm{s} \tag{10.9}$$

where

h = Planck's constant, J s
k = Boltzmann's constant, J K^{-1}
T = temperature, 273.15 K

With experimental values typically between 10^{-12} s and 10^{-14} s, a value of 10^{-13} s is often used. The activation energies are material dependent and vary between 400–100,000 kJ kmol^{-1}. Typical residence times are given in Table 10.8. *Physisorption* is a process whereby a molecule adheres to a surface without the formation of a chemical bond, usually by van der Waals forces or electrostatic attraction. *Chemisorption* refers to the formation of a chemical bond that requires a higher activation energy of desorption.

The residence time is especially sensitive to temperature. This can be shown from Eq. (10.8) as

$$\left(\frac{\delta\tau_s}{\tau_s}\right) = -\frac{E_{s,a}}{RT}\left(\frac{\delta T}{T}\right) \tag{10.10}$$

For example, for an activation energy for desorption of 14,644 kJ kmol^{-1} (3.5 kcal gmol^{-1}) and a temperature of 273.15 K, Eq. (10.10) gives

Table 10.8 Typical residence times at T = 273.15 K

Outgassing	Desorption activation energy, kcal gmol^{-1}	Desorption activation energy, kJ kmol^{-1}	Residence time, s
He	0.1	418	2.0×10^{-13}
H_2 physiosorption	1.5	6,276	2.7×10^{-12}
Ar, CO, N_2, CO_2 physisorption	3.5	14,644	1.1×10^{-10}
H_2, H_2O chemisorption	20	83,680	1.7×10^{3}

$$\left(\frac{\delta\tau_s}{\tau_s}\right) = -6.4\left(\frac{\delta T}{T}\right) \tag{10.11}$$

Thus the fractional change in residence time is 6.4 times the fractional change in temperature.

In some cases, outgassing does not follow the first-order reaction of Eq. (10.5). Sometimes outgassing has a rate that is better modeled independently of the mass and as a power of the time, where

$$\frac{\mathrm{d}m_s}{\mathrm{d}t} = -k_s(t+\alpha_s)^{-n} \tag{10.12}$$

where

$$k_s = c_s\, e^{-E_{s,a}/RT} \tag{10.13}$$

and c_s, n are parameters that fit the data, and α_s is determined by the outgassing mass rate when $t = 0$.

The time dependency for glasses (that outgas H_2O and CO_2) and elastomers (that outgas CO, CO_2, H_2, and H_2O) is primarily by diffusion and is inversely proportional to the square root of time, $t^{-1/2}$, so that $n = 1/2$ in Eq. (10.12) (Scialdone 1972). Metals typically outgas CO, CO_2, H_2, H_2O, and O_2 for the first few days, and the flux is typically inversely proportional to time, t^{-1}, so that $n=1$ in Eq. (10.12) (Scialdone 1972). For $n = 1/2$, Eq. (10.12) gives

$$m_s = m_{s,0} - 2k_s(t+\alpha_s)^{1/2} + 2k_s\alpha_s^{1/2} \tag{10.14}$$

and for $n = 1$, Eq. (10.12) gives

$$m_s = m_{s,0} - k_s \log_e(t+\alpha_s) + k_s \log_e \alpha_s \tag{10.15}$$

10.6.2 Outgassing Density in Vicinity of Spacecraft

The density of the contaminants in the vicinity of the spacecraft has been analyzed by Scialdone (1972). Based on the assumptions that the spacecraft is spherical and that the contaminants that leave the spacecraft do not collide with each other but do collide with ambient molecules, the number density, pressure, and flux as a function of distance from the spacecraft is given by

$$n_D(r) = \frac{N_D}{4\pi v_D(R+x)^2}\exp\left(-\frac{v_D+v_0}{v_D}\frac{x}{\lambda_0}\right)_s \tag{10.16}$$

$$p_D(r) = n_D kT = \frac{N_D kT}{4\pi v_D(R+x)^2}\exp\left(-\frac{v_D+v_0}{v_D}\frac{x}{\lambda_0}\right) \tag{10.17}$$

$$\phi_D(r) = n_D v_D = \frac{N_D}{4\pi(R+x)^2} \exp\left(-\frac{v_D + v_0}{v_D}\frac{x}{\lambda_0}\right) \tag{10.18}$$

where

$n_D(r)$ = number density of molecular contaminants, molecules m^{-3}
N_D = number of molecular contaminants of species s released per unit time, molecules s^{-1}
$p_D(x)$ = pressure, N m^{-2}
R = radius of the spacecraft, m
x = radial distance from surface of spacecraft, m
λ_0 = mean free path of ambient molecules, m
v_0 = spacecraft velocity, m s^{-1}
v_D = thermal velocity of desorbed molecules of species s, m s^{-1}
$\phi_D(r)$ = flux of molecular contaminants of species s, molecules m^{-2} s^{-1}

The pressure is given by $p_D = n_D kT$, and the flux is given by $\phi_D = n_D v_D$. With $v_D = 400$ m s^{-1}, and both the mean free path λ_0 and spacecraft velocity v_0 a function of altitude, parametric plots of Eqs. (10.16) and (10.17) are given in Fig. 10.3.

The ratio of the flux of molecular density, pressure, and flux of the returning molecular contaminants to the flux leaving the surface is also given by Scialdone (1972) as

$$\frac{n_{D,r}}{n_D} = \frac{p_{D,r}}{p_D} = \frac{R v_D\left(\frac{v_0}{v_D}+1\right)}{\lambda_0 v_{D,r}} \tag{10.19}$$

$$\frac{\phi_{D,r}}{\phi_D} = \frac{R}{\lambda_0}\left(\frac{v_0}{v_D}+1\right) \tag{10.20}$$

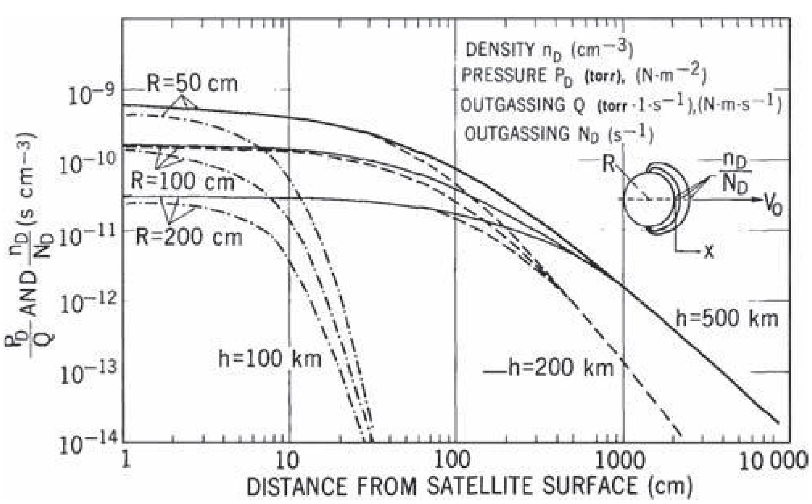

Fig. 10.3 Density and pressure from outgassing as a function of distance from spherical spacecraft. [From Scialdone (1972); courtesy of NASA.]

where

$n_{D,r}$ = density of returning molecular contaminants, molecules m^{-3}
$p_{D,r}$ = pressure of returning molecular contaminants, N m^{-2}
R = radius of the spacecraft, m
$v_{D,r}$ = velocity of returning molecular contaminants, m s^{-1}
$\phi_{D,r}$ = flux of returning molecular contaminants, molecules m^{-2} s^{-1}

The flux distribution of outgassing material escaping from a vent is often modeled by

$$\phi(r,\theta) = \frac{n+1}{2\pi r^2}\frac{\mathrm{d}m_v}{\mathrm{d}t}\cos^n\theta \tag{10.21}$$

where

$\mathrm{d}m_v/\mathrm{d}t$ = mass flow through a vent, kg s^{-1}
n = exponent, $1 \le n \le 2.5$, often taken as 2
r = radius from vent, m
t = time, s
θ = angle with respect to vent axis
$\phi(r, \theta)$ = flux, kg s^{-1} m^{-2}

10.6.3 Contamination Analysis Software

Several software programs are available to analyze the potential for contamination of spacecraft in orbit; see ESA (2006) and Space Environment Information System (2007) for detailed discussions. Models available include the following. CONTAM is a digital simulation used for computing the plumes of propulsion systems (Hoffman et al. 1985). TRICONTAM is a program devoted to the contamination from thruster plumes (Trinks 1991a). The Spacecraft/Orbiter Contamination Representation Accounting for Transiently Emitted Species (SOCRATES) is a Monte Carlo simulation of the products of contamination (Trinks 1991b). The Shuttle/Payload Contamination Evaluation code (SPACE II) is a code to assess contamination of the shuttle and spacelab (Jarossy 1981). Molecular Flux (MOLFLUX) was used to predict contamination of the International Space Station (Babel et al. 1996).

10.7 Contamination Assessments

10.7.1 Introduction

Reductions and assessment of contamination from outgassing can be accomplished by subjecting the flight hardware to a thermal vacuum environment during which a vacuum is maintained while the temperature is cycled between hot and cold extremes. Vacuum baking can significantly reduce the outgassing observed in orbit. Several inspection techniques can be used to verify the level of cleanliness of flight hardware during the different phases of the thermal vacuum test, while in the clean room, and during other testing prior to launch. Particulates can be measured by visual inspection using intense light, optical microscopy, temperature quartz crystal microbalances, bidirectional reflectance distribution function measurements, and tape lifts. Cold fingers, scavenger plates, and witness mirrors are surfaces at low temperatures that can be used to collect contaminants for assessment. Composition can be determined by residual gas analyzers, infrared spectrometers, gas chromatographs, and mass spectrometers.

10.7.2 Temperature Quartz Crystal Microbalances

Temperature quartz crystal microbalances (TQCM) are used to monitor contamination of critical surfaces during test, launch, and in orbit. TQCMs are piezoelectric devices consisting of a thin wafer of quartz with electrodes affixed to each side whose frequency upon excitation is a function of its mass. They are capable of measuring nanograms of changes in mass. The mass flux at the TQCM is given by

$$\dot{m} = S_t A_{\mathrm{TQCM}} \frac{\mathrm{d}f}{\mathrm{d}t} \tag{10.22}$$

where the usual units used are

A_{TQCM} = surface area of the TQCM, cm^2
$\mathrm{d}f/\mathrm{d}t$ = change in frequency, Hz s^{-1}
$\dot{m}$ = mass flux, g s^{-1}
S_t = TQCM sensitivity g cm^{-2} Hz^{-1}

10.7.3 Tape Lifts

The tape lift method employs a section of 3M 480 polyethylene tape with acrylic adhesive placed onto the surface to be tested and then removed and placed on a clean microscope slide. The tape backing is dissolved and the particulates are fixed with a mounting medium that is then covered to trap the population. A microscope is used to count the number of particulates according to size. The standard ASTM E1216-06 (American Society for Testing and Materials 2006) gives procedures for sampling surfaces to determine the presence of particulate contamination, 5 μm and larger by tape lift. Molecular contamination levels can be determined by analyzing a solvent wash using infrared or mass spectrometry. The measurements can be used to verify that the number of contaminants is within the specified limits and to provide information to help identify the source and location of contamination.

10.7.4 Bidirectional Reflectance Distribution Function

The bidirectional reflectance distribution function (BRDF) is the angular distribution of light scattered from a surface as a function of wavelength, polarization state, and angle of incidence. The BRDF characterization of scatter is an important technique to characterize surface properties of optical systems and the effects of the contamination of surfaces. Measurement of contamination mass alone, as with a TQCM, does not give the effect of the contamination on the properties of the electromagnetic radiation on the contaminated surface. For example, a given amount of contaminant deposited uniformly on a surface will have significantly different optical properties compared with the same amount of contaminant deposited as small droplets. The latter can produce significantly more scattering over a wider range of wavelengths. The BRDF is a more direct measurement of real-time changes to optical properties due to deposition of contaminants.

Spectral radiance, L_λ, is defined as the power of electromagnetic radiation emitted or reflected from a surface per unit solid angle per unit of projected area per wavelength. *Spectral irradiance*, E_λ, is defined as the power per unit area per unit wavelength of electromagnetic radiation incident on a surface and is equal to the component of electromagnetic radiation normal to the surface integrated over the hemisphere. The relationship between radiance and irradiance, using Fig. 10.4, is

$$\mathrm{d}E_\lambda(\theta_i,\phi_i,\lambda) = \frac{\mathrm{d}A}{r^2} L_\lambda\ (\theta_i,\phi_i,\lambda),\ \cos\theta_i = L_\lambda\ (\theta_i,\phi_i,\lambda)\cos\theta_i\,\mathrm{d}\Omega_i \tag{10.23}$$

where

$\mathrm{d}A$ = differential area, m^2
E_λ = spectral irradiance, W m^{-3}
L_λ = spectral radiance, W m^{-3} sr^{-1}
r = distance between the surface and the differential area, m
θ_i = polar angle of the incident electromagnetic radiation
ϕ_i = azimuth angle of the incident radiation
$\mathrm{d}\Omega_i$, $\mathrm{d}\Omega_r$ = solid angle of incident and reflected electromagnetic radiation, sr

The BRDF is the ratio of the reflected radiance of electromagnetic radiation to the incident irradiance at a particular wavelength. If a surface is a perfect mirror,

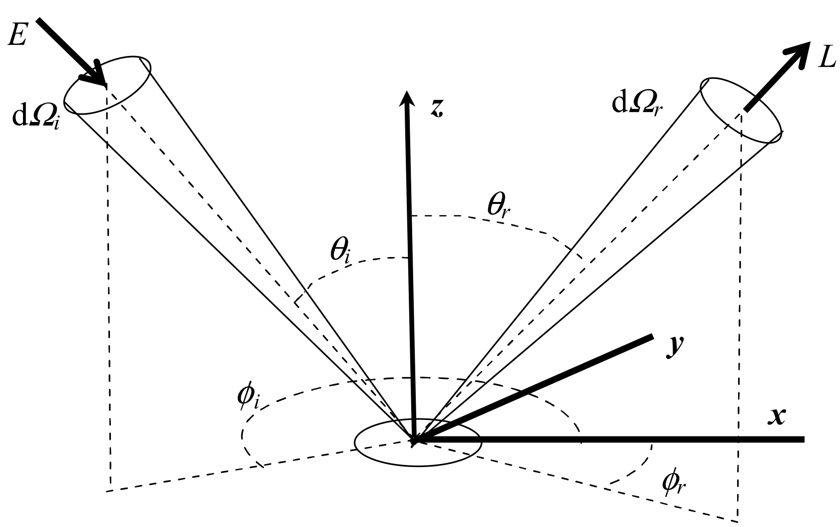

Fig. 10.4 Geometry for BRDF.

the BRDF is zero in all directions except in the direction of the specular reflection. The integral of BRDF over the hemisphere is the *hemispherical reflectance*. Low values of BRDF outside of the specular reflection region indicate a surface free from scattering centers. Contributors to BRDF are surface roughness, molecular and particulate contamination, and surface coatings. The BRDF is defined by

$$f_r(\theta_i,\phi_i,\theta_r,\phi_r,\lambda) = \frac{\mathrm{d}L_\lambda(\theta_i,\phi_i,\theta_r,\phi_r,\lambda)}{\mathrm{d}E_\lambda(\theta_i,\phi_i,\lambda)} = \frac{\mathrm{d}L_\lambda(\theta_i,\phi_i,\theta_r,\phi_r,\lambda)}{L_\lambda(\theta_i,\phi_i,\lambda)\cos\theta_i \mathrm{d}\Omega_i} \tag{10.24}$$

where

$f_r(\theta_i, \phi_i, \theta_r, \phi_r, \lambda)$ = BRDF, sr^{-1}
θ_i, ϕ_i = spherical coordinates of incoming radiance
θ_r, ϕ_r = spherical coordinates of reflected radiance
λ = wavelength of electromagnetic radiation
Ω_i, Ω_r = solid angle of incident and reflected radiation, sr

The amount of light reflected from the surface in a given direction follows from Eq. (10.24) as

$$L_r(\theta_i,\phi_i,\theta_r,\phi_r,\lambda) = \int_{\Omega_r} f_r(\theta_i,\phi_i,\theta_r,\phi_r,\lambda)\ L_\lambda(\theta_i,\phi_i,\lambda)\cos\theta_i\,\mathrm{d}\Omega_i \tag{10.25}$$

Since the amount of light reflected must be equal to or less than the incident light, then

$$\int_{\Omega_r} f_r(\theta_i,\phi_i,\theta_r,\phi_r,\lambda)\cos\theta_i\,\mathrm{d}\Omega_r \le 1 \tag{10.26}$$

For a single point-light source, the light reflected is given by

$$L_\lambda(\theta_i,\phi_i,\theta_r,\phi_r,\lambda) = f_r(\theta_i,\phi_i,\theta_r,\phi_r,\lambda) L_\lambda(\theta_i,\phi_i,\lambda)\cos\theta_i \tag{10.27}$$

Various models have been proposed for the BRDF to account for surface characteristics. Often the models decompose the reflected power into three components: ideal specular reflection, ideal diffuse reflection, and directional diffuse reflection.

One of the simplest BRDF was proposed by Lambert (1760) and has the constant value

$$f_{r,\mathrm{Lambertian}}(\theta_i,\phi_i,\theta_r,\phi_r,\lambda) = \frac{\rho_d}{\pi} \tag{10.28}$$

where

ρ_d = diffuse reflectivity ≤ 1

Lambert's BRDF describes the reflection properties of diffuse reflection as a cosine relationship as seen from Eq. (10.27).

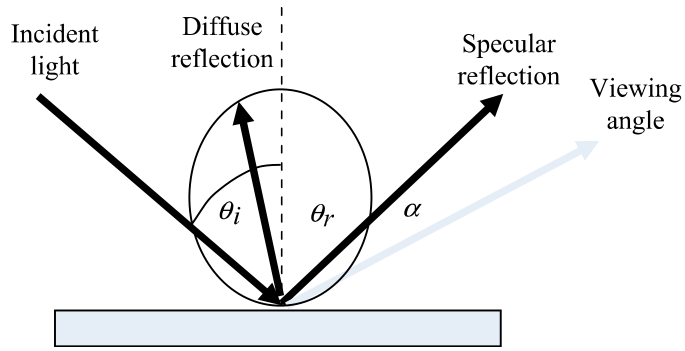

Fig. 10.5 Geometry for modified Phong BRDF.

A simple heuristic model that demonstrates the properties of a surface is the modified Phong (1975) model that separates the reflection power distribution from a surface into diffuse and specular components, as illustrated in Fig. 10.5:

$$f_r(\theta_i,\phi_i,\theta_r,\phi_r,\lambda) = f_{r,d}(\theta_i,\phi_i,\theta_r,\phi_r,\lambda) + f_{r,s}(\theta_i,\phi_i,\theta_r,\phi_r,\lambda)$$
$$= \frac{\rho_d}{\pi} + \frac{\rho_s(n+2)\cos^n\alpha}{2\pi} \tag{10.29}$$

where

$f_{r,d}(\theta_i, \phi_i, \theta_r, \phi_r, \lambda)$ = diffuse component of the BRDF
$f_{r,s}(\theta_i, \phi_i, \theta_r, \phi_r, \lambda)$ = specular component of the BRDF
n = specular exponent, dimensionless
α = viewing angle from the specular reflective peak, $-\frac{\pi}{2} \leq \alpha = \frac{\pi}{2}$
$\theta = \theta_i = \theta_r$ = angle from the normal to the surface
ρ_d = diffuse reflectivity, fraction of the incoming power that is reflected diffusely
ρ_s = specular reflectivity, fraction of the perpendicular incoming power reflected specularly
and $\rho_d + \rho_s \leq 1$ for power conservation.

Larger values of n represent an increasingly smoother surface with a concentration of the reflection in the vicinity of the specular reflection direction. Examples of the BRDF for the modified Phong model are illustrated in Fig. 10.6.

10.8 Planetary Protection

Planetary biological protection is a policy to prevent the contamination of other celestial bodies by biological agents foreign to them and prevent the contamination of the Earth by extraterrestrial biological agents. Protection of extraterrestrial bodies from biological and organic contamination (known as avoiding forward contamination) would preserve the opportunity to explore fully their organic and biological chemistry. Although the existence of life on extraterrestrial bodies is unlikely, it is important not to jeopardize scientific investigations on the precursors of extraterrestrial life forms. Protection of contamination of the Earth from

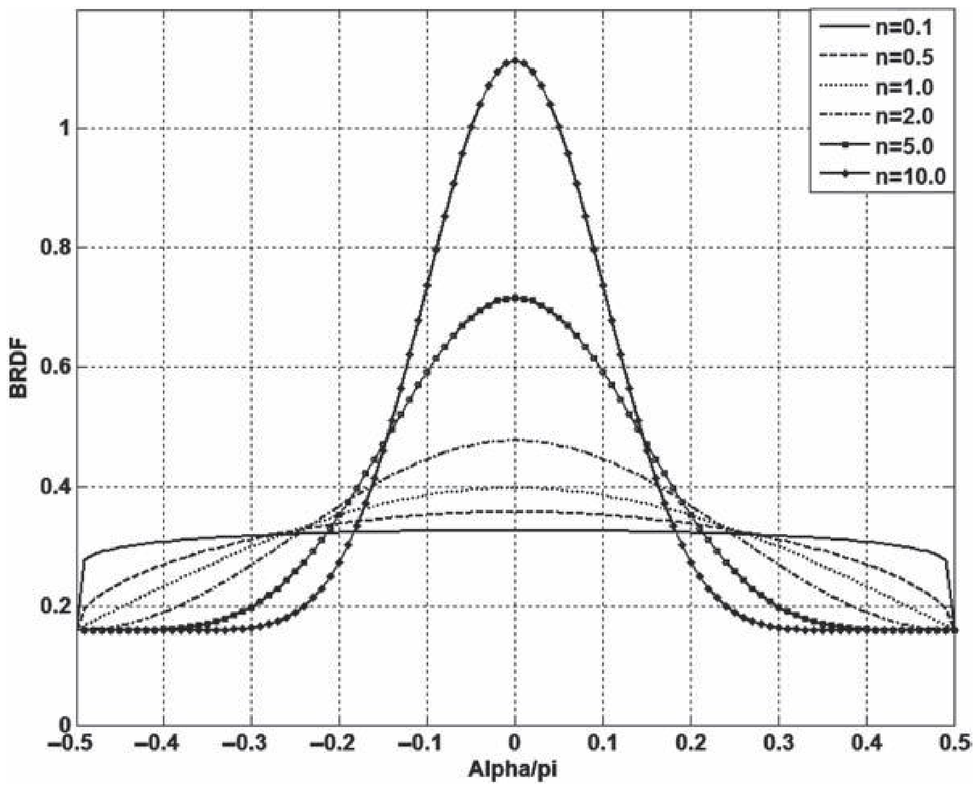

Fig. 10.6 BRDF modified Phong model for $\rho_d = \rho_s = 0.5$.

extraterrestrial biological and organic agents (known as avoiding backward contamination) is to protect the Earth's ecological system that includes plants, animals, and humans from potential hazards.

NASA's policy for planetary protection is defined in NASA Policy Directive (NPD) 8020.7F (NASA 1999). The planetary protection policy defined by the Committee on Space Research (COSPAR) of the International Council of Scientific Unions (ICSU) (Committee on Space Research 2005) and the NASA specific mission requirements, given in NASA Procedural Requirement (NPR) 8020.12C (NASA 2005), are consistent and are summarized in Table 10.9, where five categories of missions are defined.

Examples of specific mission requirements to minimize forward contamination for selected missions described in NASA NPR 8020.12C (NASA 2005) are given in Table 10.10.

To satisfy the biological requirements imposed on robotic missions in categories II to V, a biological decontamination program and a biological assessment program to certify that the requirements are met are required. Procedures for assessing the microbial status of space hardware are defined in NASA NPG 5340.1D (NASA 2006). The techniques used to reduce onboard microorganisms to the recommended numbers include cleaning surfaces with alcohol, hydrogen peroxide vapor, ultraviolet radiation, radiation, dry heat and then integrating the instrument or spacecraft in a clean room. Sterilization by heat is accomplished at a temperature of 125°C for 5 h in an arid environment. This technique can be used for items such as thermal blankets, honeycomb structures, parachutes, etc. The clean room used is generally ISO class 8 or lower.

Table 10.9 Mission categories and protection characteristics for robotic missions

Mission categories	Mission risk	Mission types	Requirements
I	Missions not interested in process of chemical evolution or origin of life with no chance to jeopardize future exploration	Flyby; Orbiter; Lander: Earth's moon, Mercury, undifferentiated metamorphosed asteroids	1. Certification of category I mission, only
II	Missions with significant interest in process of chemical evolution or origin of life with remote chance to jeopardize future exploration	Flyby, Orbiter, Lander: Venus, Jupiter (exclusive of icy moons), Saturn, Titan, Uranus, Neptune, Triton, Pluto/Charon; Kuiper-belt objects, comets, carbonaceous chrondrite asteroids	1. Planetary protection plan 2. Prelaunch planetary protection report 3. Postlaunch planetary protection report 4. End-of-mission report
III	Missions (mostly flyby and orbiter) to a target body of chemical evolution and/or origin of life interest or for which scientific opinion provides a significant chance of contamination that could jeopardize future biological experiments	Flyby, Orbiter: Mars, Europa, Ganymede, Callisto	1. Planetary protection plan 2. Prelaunch planetary protection report 3. Postlaunch planetary protection report 4. Organics inventory 5. End-of-mission report
IV	Missions (mostly probes and landers) to a target body of chemical evolution and/or origin of life interest or for which scientific opinion provides a significant chance of contamination that could jeopardize future biological experiments	Lander, Probe: Mars, Europa, Ganymede, Callisto	1. Planetary protection plan 2. Prelaunch planetary protection report 3. Postlaunch planetary protection report 4. Organics inventory 5. End-of-mission report
V	Protection of the terrestrial system, the Earth and the moon	Unrestricted Earth return	1. Documentation for outbound phase 2. Certification of unrestricted Earth return
		Restricted Earth return	1. Planetary protection plan including outbound phase and Earth safety analysis plan 2. Prelaunch planetary protection report 3. Postlaunch planetary protection report 4. Earth Preentry report 5. Sample Prerelease report 6. End-of-mission report

Source: From NASA NPR 8020.12C (NASA 2005) and COSPAR (Committee on Space Research 2005); courtesy of NASA.

Table 10.10 Selected spacecraft missions biological requirements

Title	Definition	Source	Value			Application		
			Upper	Acceptable	Lower	Mission	Category	Planet
Orbit lifetime probability	Maximum probability of impact of Mars by an orbiter or any subsystem over a specified lifetime			Spacecraft will remain in orbit between 20 to 50 years with probability of >0.95				Mars
Maximum probability of accidental impact	Maximum probability of accidental impact		0.01	0.01	0.01	Orbiter Flyby	III	Mars
Maximum total microbial spore burden	Maximum total microbial spore burden including all subsystems	Mars orbiter not meeting lifetime requirement		$\leq 5.0 \times 10^5$ spores per vehicle		Orbiter	III	Mars
Maximum surface microbial spore burden	Maximum limit of exposed surface bioburden			≤ 300 bacterial spores m^{-2} $\leq 5.0 \times 10^5$ spores per vehicle		Lander, Probe	IVa (Landers not carrying instruments for investigation of extant Martian life)	Mars
Maximum surface microbial spore burden	Establishes a maximum limit on the free surface	All free exterior and interior spacecraft surfaces		≤ 30 bacterial spores on the free surfaces on a landed system		Lander, Probe	IVb (Landers carrying instruments for investigation of extant Martian life) IVc (Landers that investigate special regions)	Mars
Maximum probability of contamination of a Europan ocean	Maximum probability of contamination of a Europan ocean with terrestrial contamination	Europa flybys, orbiters, landers, and probes		Probability of contamination should be $\leq 1.0 \times 10^{-4}$		All	III and IV	Europa

Source: From NASA NPR 8020.12C (NASA 2005); courtesy of NASA.

The bioburden is assessed by collection of a sample with a cotton swab and counting the number of spores formed in Petri dishes, as described in NASA NPG 5340.1D (NASA 2006). A *spore* is a small, usually single-celled reproductive organism that is generally resistant to heat, dryness, and radiation and is capable of growing into a new organism, produced especially by some bacteria, fungi, and algae.

References

American Society for Testing and Materials (2003a), "Standard Test Method for Total Mass Loss and Collected Volatile Condensable Materials from Outgassing in a Vacuum Environment," ASTM E-595-93(2003)e2, West Conshohocken, PA.

American Society for Testing and Materials (2003b), "Standard Test Method for Contamination Outgassing Characteristics of Spacecraft Materials," ASTM E1559-03, West Conshohocken, PA.

American Society for Testing and Materials (2006), "Standard Practice for Sampling for Particulate Contamination by Tape Lift," ASTM E1216-06, West Conshohocken, PA.

Babel, H., Hasegawa, M., Jones, C., and Fussell, J. (1996), "The Effects of Contamination from Silicones and a Modified-Tefzel Insulation," IAF Paper 96-I.5.08, Oct.

Campbell, W. A., Jr., and Scialdone, J. J. (2006), "Outgassing Data for Selecting Spacecraft Materials," NASA RP-1124, NASA Center for AeroSpace Information, Linthicum Heights, **MD, updated 21 April 2006; available online at http://outgassing.nasa.gov/.**

Committee on Space Research (2005), "Planetary Protection Policy," International Council of Scientific Unions, Paris, amended 24 March.

U.S. Department of Defense (1962), "Product Cleanliness Levels and Contamination Control Program," MIL-STD-1246, U.S. Government Printing Office, Washington, DC.

ESA (2006), "ECSS Space Environment Standard," ECSS E-10-04, European Space Research and Technology Centre, Noordwijk, The Netherlands.

Hoffman, R. J., Kawasaki, A., Trinks, H., Bindemann, I., and Ewering, W. (1985), "The CONTAM 3.2 Plume Flowfield Analysis and Contamination Prediction Computer Program: Analysis Model and Experimental Verification," AIAA Paper 85-0928, June.

Institute of Environmental Sciences and Technology (2002), "Product Cleanliness Levels and Contamination Control Program," IEST-STD-CC1246D, formerly MIL-STD-1246, Rolling Meadows, IL.

International Organization for Standardization (1999), "Cleanrooms and Associated Controlled Environments—Part 1: Classification of Air Cleanliness," ISO-14644-1, Geneva.

Jarossy, F. J., Pizzicaroli, L. C., and Owen, N. L. (1981), Shuttle/Payload Contamination Evaluation (SPACE) Program Improvements," *Shuttle Optical Environment; Proceedings of the Meeting*, Washington, DC, April 23–24, pp. 78–85.

Lambert, J. H. (1760), *Photometria sive de mensura et grabibus luminis, colorum et umbrae*, Eberhard Klett, Augsberg, Germany.

NASA (1999), "Biological Contamination Control for Outbound and Inbound Planetary Spacecraft," NASA NPD 8020.7F, Washington, DC, 19 Feb., revalidated 23 Oct. 2003.

NASA (2005), "Planetary Protection Provisions for Robotic Extraterrestrial Missions," NASA NPR 8020.12C, Washington, DC, 27 April.

NASA (2006), "NASA Standard Procedures for the Microbial Examination of Space Hardware," NPG 5340.1D, Washington, DC.

NASA (undated), "NASA Preffered Reliability Practices, Contamination Control Program," PD-ED-1233, Goddard Space Flight Center, Greenbelt, MD.

Phong, B. T. (1975), "Illumination for Computer Generated Images," *Communications of the ACM*, Vol. 18, No. 6, pp. 311–317.

Scialdone, J. J. (1972), "Self Contamination and Environment of an Orbiting Spacecraft," NASA TN D-6645, May.

Space Environment Information System (2007), Belgian Institute for Space Aeronomy, ESA, ECSS Secretariat, European Space Research and Technology Centre, Noordwijk, The Netherlands, http://www.spenvis.oma.be/spenvis/ecss/annexi/annexi.html.

Trinks, H. (1991a), "Exhaust Plume Data Handbook (EPDH IV), Progress Report IV," ESA Contract No. 7510/87/NL/PP, Sept.

Trinks, H. (1991b), "Surface Effect Data Handbook (SEDH III), Progress Report V," ESA Contract No. 7510/87/NL/PH, Sept.

Problems

10.1 Find an article in a peer-reviewed journal on spacecraft contamination and summarize it in three pages.

10.2 Find an article in a peer-reviewed journal on quartz crystal microbalances and summarize its detailed operation in three pages.

10.3 The bidirectional reflectance distribution function (BRDF) for specular reflection of surfaces is given by the Phong model to be proportional to $\cos^n \alpha$ where $\theta = \theta_i = \theta_r$ is the angle from the vertical, and α is the angle measured from the specular reflection direction as in Fig. 10.5. Plot the response for $n = 0.1, 0.5, 1.0, 2.0, 5.0, 10.0$, over the range $-\pi/2 \leq \alpha \leq \pi/2$.

10.4 The bidirectional reflectance distribution function (BRDF) for specular reflection of surfaces is given by the Phong model for specular reflection to be $f_r(\theta_i, \phi_i, \theta_r, \phi_r, \lambda) = \cos^n \alpha$ where $\theta = \theta_i = \theta_r$ is the angle from the vertical and α is the angle measured from the specular reflection direction as in Fig. 10.5. Plot the response for $n = 1, 10, 50, 100, 200$, over the range $-\pi/2 \leq \alpha \leq \pi/2$.

10.5 Show that the constant Lambert BRDF must be $\leq \pi^{-1}$ if the amount of reflected light is not to be greater than the incident light.

10.6 Show for the outgassing of a metal that the coefficients in the equation for the mass loss, Eq. (10.12), $\dot{m}_s = -k_s(t + \alpha_s)^{-n}$ for $n = 1$ can be determined for two measurements of $\dot{m}_s$ as

$$\alpha = \frac{Rt_1 - t_2}{1 - R}$$